

24/24

Excellent!

100%

PHY201 Final Exam on chapter(s): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

Name: _____

These questions assess your understanding of the material from Chapter(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. Two billiard balls, specifically the ten-ball and the twelve-ball, move toward each other until they elastically collide. The twelve-ball begins at rest until a force of 43 N is applied to it for a brief 12 ms, at which point it is moving toward the ten-ball. Calculate the velocity of the ten-ball after it elastically collides with the twelve-ball, given that each ball has the same mass of 164 g and the velocity of the ten-ball is 3.3 m/s, moving toward the twelve-ball.

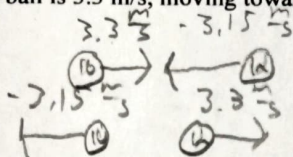
$$F_{avg} \Delta t = m \Delta v$$

$$\therefore v = \frac{F_{avg} \Delta t}{m}$$

$$v = \frac{(43)(0.012)}{0.164}$$

$$v = 3.15 \frac{m}{s}$$

Because it is traveling in the negative x-direction



ANSWER: _____

$$v_{10i} = v_{10f}$$

$$v_{10i} = v_{10f}$$

$$\therefore v_{10f} = -3.15 \frac{m}{s}$$

Both is an elastic collision between two object with the same mass so velocities swapped

2. Stephen and Paul are playing rugby. Stephen pushes Paul with a force of 694 N. Stephen has a mass of 113 kg while Paul has a mass of 120 kg. What is Paul's acceleration as a result of getting hit by Stephen?

ANSWER: _____

1

3. The aorta pumps blood from the heart to the rest of the body. Calculate the average speed of the blood in Ian's aorta if the flow rate of his blood is 8.2 L/min. Ian's aorta has a radius of 12 mm.

$$Q_a = v_a \cdot A_a$$

$$A = \pi \cdot r^2 \quad r = 12 \text{ mm} \rightarrow 0.012 \text{ m}$$

$$A = \pi \cdot (0.012)^2$$

$$A = 4.5238 \times 10^{-4}$$

$$\therefore Q = v_a \cdot A_a$$

$$\therefore v_a = \frac{Q}{A}$$

ANSWER: 0.302 $\frac{m}{s}$

$$8.2 \frac{L}{min} \left| \frac{1m^3}{1000L} \frac{1min}{60s} \right| = 1.367 \times 10^{-4}$$

$$\therefore v_a = \frac{1.367 \times 10^{-4}}{4.5238 \times 10^{-4}}$$

$$v_a = 0.302 \frac{m}{s}$$

+2

4. Scott puts a 0.65-kg aluminum pot on the stove and fills it with 0.32 liters of water. Both the pot and the water are initially at 13.1°C. How much heat is required to bring them both to 71.1°C? The specific heat value of water is 4186 J/kg°C and for aluminum it is 900 J/kg°C.

ANSWER: 111622.16 J

$$Q_{\text{total}} = Q_{\text{pot}} + Q_{\text{water}}$$

$$Q = m_{\text{pot}} c \Delta T$$

$$Q = (0.65)(900)(58)$$

$$Q_p = 33930$$

$$Q = m_{\text{water}} c \Delta T$$

$$Q = (0.32)(4186)(58)$$

$$Q_w = 77692.16$$

$$\Delta T = T_f - T_i$$

$$\Delta T = 71.1 - 13.1$$

$$\Delta T = 58^\circ \text{C}$$

$$\therefore \frac{1 \text{ L}}{1 \text{ kg}} \therefore \frac{0.32}{1 \text{ kg}} = 0.32 \text{ kg}$$

$$Q_{\text{total}} = 33930 + 77692.16 = 111622.16 \text{ J}$$

5. Ethan drives his 1000-kg car around a 240-m radius unbanked curve at 26.3 m/s (about 58 mph). Ethan's mass is 85 kg. Calculate the centripetal force exerted on the car as Ethan drives around the curve.

ANSWER: 3127.015 N

$$F_c = m \frac{v^2}{r} \quad \therefore F_c = m r \omega^2$$

$$\therefore F_c = (1000 + 85) \cdot \left(\frac{26.3}{240} \right)^2$$

$$F_c = 3127.015 \text{ N}$$

6. William creates a soap bubble and measures its diameter to be 116 μm. Calculate the absolute pressure inside of the bubble, given that the surface tension of typical soapy water is 0.0370 N/m. Report your answer in mmHg, recalling that 133.32 Pa is 1 mmHg.

ANSWER: 779.15 mmHg

$$116 \mu\text{m} \rightarrow \frac{116 \times 10^{-6} \text{ m}}{2}$$

$$P_g = \frac{4\gamma}{r} \quad r = 5.8 \times 10^{-5}$$

$$P_g = \frac{4(0.0370)}{5.8 \times 10^{-5}}$$

$$P_g = 2551.22$$

$$P = P_g + P_{\text{atm}} \quad \therefore 101325 \text{ Pa}$$

$$\therefore P_{\text{abs}} = 2551.22 + 101325$$

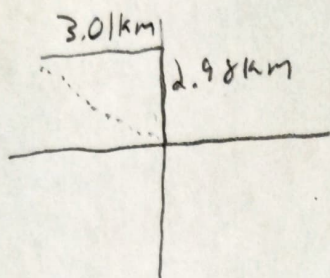
$$P_{\text{abs}} = 103876.22$$

$$\therefore 1 \text{ mmHg} = 133.32 \text{ Pa}$$

$$\therefore \frac{103876.22}{133.32} = 779.15 \text{ mmHg}$$

+3

7. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Shiv's. To get to Shiv's, you have to walk 3.01 km west and then 2.98 km north. How far does he live from you if you drew a straight line on the map?



$$c = \sqrt{a^2 + b^2}$$

$$c = \sqrt{2.98^2 + 3.01^2}$$

$$c = 4.24 \text{ km}$$

ANSWER: 4.24 km



8. A rocket with a mass at liftoff of 2.7 million kg burns fuel at a rate of 17100 kg/s. Its exhaust velocity was measured to be 2800 m/s. Calculate its acceleration at liftoff.

$$a_r = \frac{v_e \dot{m}}{m \Delta t} - g$$

$$a_r = \frac{17100 \cdot 2800}{2.7 \times 10^6} - 9.8$$

$$a_r = 7.93 \frac{\text{m}}{\text{s}^2}$$

ANSWER: 7.93 $\frac{\text{m}}{\text{s}^2}$



9. Shiv kicks a soccer ball, initially at rest on the ground, with an initial velocity of 8.9 m/s at a launch angle of 21.8° above the horizontal. The ball flies through the air and lands back on the ground. Calculate its total flight time, assuming that air resistance is negligible.

$$z_{iy} = 0$$

$$z_y = z \sin \theta$$

$$z_y = 8.9 \sin 21.8$$

$$z_y = 3.305 \frac{\text{m}}{\text{s}}$$

ANSWER: .624 s

$$v_{fx} \text{ at peak} = 0$$

$$v_x = v_x + a t$$

$$0 = 3.305 - 9.8 t$$

$$\frac{9.8 t}{9.8} = \frac{3.305}{9.8}$$

$$t = .333 \text{ s}$$

$$\text{time}_{\text{total}} = 2(.333) = .624 \text{ s}$$

+3

10. As Jonathan waits for his food to heat up, he determines that his microwave oven rotates the food with a speed of 3.6 revolutions/minute. What is the food's angular velocity in radians per second?

$$3.6 \frac{\text{rev}}{\text{min}} \left(\frac{1 \text{ min}}{60 \text{ s}} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = 0.376 \frac{\text{rad}}{\text{s}}$$

ANSWER: 0.376 ^{rad}/_s

11. Scott attaches a nozzle of diameter 6.2 mm to a garden hose of diameter 1.6 cm. If the hose fills up a 3-L bucket in 14.3 seconds, calculate the pressure in the hose assuming level, frictionless flow.

$$Q_n = A_n \cdot v_n$$

$$A_n = \pi r^2$$

$$A_n = \pi (3.1 \times 10^{-3})^2$$

$$A_n = 3.014 \times 10^{-5}$$

$$\therefore v_n = \frac{Q}{A_n}$$

$$\therefore v_n = \frac{2.042 \times 10^{-4}}{3.014 \times 10^{-5}}$$

$$v_n = 6.95 \frac{\text{m}}{\text{s}}$$

$$Q_h = A_h \cdot v_h$$

$$A_h = \pi r^2$$

$$A_h = \pi (1.6 \times 10^{-2})^2$$

$$A_h = 1.0106 \times 10^{-4}$$

$$\therefore v_h = \frac{Q}{A_h}$$

$$\therefore v_h = \frac{2.042 \times 10^{-4}}{1.0106 \times 10^{-4}}$$

$$v_h = 1.04 \frac{\text{m}}{\text{s}}$$

ANSWER: 124935.45 Pa (Pa)

$$Q = \frac{V}{t} = \frac{3 \times 10^{-3}}{14.3} = 2.097 \times 10^{-4}$$

$$P_h + \frac{1}{2} \rho v_h^2 = P_n + \frac{1}{2} \rho v_n^2$$

The nozzle and hose are at the same height, they cancel out. $P_h = P_n$

$$\therefore P_h = P_n + \frac{1}{2} \rho (v_n^2 - v_h^2)$$

$$\therefore P_h = (1.01325) + \frac{1}{2} (1000) (6.95^2 - 1.04^2)$$

$$P_h = 124935.45 \text{ Pa}$$

12. Dylan places an outdoor sandwich-board advertising sign on the ground. The sign has a taut chain 0.44 m vertically below its hinge. The entire sign's mass is 7 kg. The sign has a vertical height of 1.29 m. The hinge is opened so that the feet of the sign measure 2.1 m apart. Calculate the magnitude of the tension force that the chain exerts on each of the sign's sides, assuming there is no friction between the feet and the sidewalk. (Drawing tip: the sign resembles a large A. The chain resembles the horizontal line in the A and the hinge is at the top of the A.)

$$\text{Vertical} = .44 \text{ m}$$

$$\text{Horizontal} = 2.1 \text{ m}$$

$$\frac{2.1}{2} = 1.05 \text{ Horizontal distance to foot}$$

$$\frac{1.05}{2} \text{ center of mass from hinge} = .525$$

$$W = mg = (7)(9.8) = 68.6 \text{ N weight of sign}$$

$$\frac{68.6}{2} = 34.3 \text{ N}$$

weight on one leg

+2

ANSWER: 40.926 N (correct)

$$\begin{cases} \sum F_{x} = 0 \\ \sum F_{y} = 0 \\ \sum \tau = 0 \end{cases}$$

$$\begin{cases} \sum F_{x} = 0 \\ \sum F_{y} = 0 \\ \sum \tau = 0 \end{cases}$$

$$\begin{cases} \sum F_{x} = 0 \\ \sum F_{y} = 0 \\ \sum \tau = 0 \end{cases}$$

$$\therefore H(0.44) = (34.3)(1.29)$$

$$H = 40.926$$

13. On a family trip, Jeremy and Isabella take their son, Adam, to visit the deepest man-made hole on Earth, the Kola Superdeep Borehole. Adam shows little interest, however, and spends all his time on his cell phone. Jeremy, fairly horrified at his son's lack of scientific interest, grabs the phone and throws it down the hole, which is 9 inches in diameter and 12,262 meters deep. Jeremy manages to hear over Adam's cries that it hit the bottom in 51.1 seconds. What was the velocity at which Jeremy threw the cell phone?

$$\begin{aligned}
 Dy &= v_0 t + \frac{1}{2} g t^2 \\
 -12,262 &= v_0(51.1) + \frac{1}{2}(-9.8)(51.1)^2 \\
 -12,262 &= v_0(51.1) - 12,746.927 \\
 \frac{532.427}{51.1} &= \frac{v_0(51.1)}{51.1} \\
 v_0 &= 10.43 \frac{m}{s}
 \end{aligned}$$

ANSWER: 10.43 $\frac{m}{s}$ ✓

14. Suppose that Benjamin floats in freshwater with 90.2% of his volume submerged when his lungs are full of air. His mass is 90 kg. What is his average density?

$$\begin{aligned}
 \rho_{\text{density}} &= \rho_{\text{water}} \times (\text{volume submerged}) \\
 \rho_d &= 1000 \times .902 \\
 \rho_d &= 902 \frac{kg}{m^3}
 \end{aligned}$$

ANSWER: 902 $\frac{kg}{m^3}$ ✓

15. Tyler is decelerating as he snowboards at 5.1 m/s up a slope that is inclined at 6.3° above the horizontal. Calculate the magnitude of his deceleration. The total mass of Tyler and his snowboard is 87 kg. The density of the air is 1.276 kg/m³. Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$. Ignore the effect of air resistance. Show all your work. The use of memorized equations absent from the equation sheet merits no credit.

ANSWER: _____

3

+2

16. Matthew flips his bicycle upside down and moves the pedals so that the rear wheel begins spinning. If the wheel starts from rest and reaches a final angular velocity of 225 rpm in 4.6 s, calculate the angular acceleration in rad/s^2 .

$$225 \text{ rpm} \left(\frac{2\pi}{1 \text{ rev}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) = 23.56 \frac{\text{rad}}{\text{s}}$$

ANSWER:

$$5.12 \frac{\text{rad}}{\text{s}^2}$$

$$\omega_f = \omega_0 + \alpha t$$

$$\frac{23.56}{4.6} = \frac{24.6}{4.6}$$

$$\alpha = 5.12 \frac{\text{rad}}{\text{s}^2}$$

17. Linda wants to use her pressure cooker to cook food in liquid water at a temperature of 157°C . To determine the pressure required inside the pressure cooker for water to boil at this temperature, she can use the Clausius-Clapeyron equation for water: $P(T) = (1 \text{ kPa}) \cdot \exp \left[(4750 \text{ K}) \left(\frac{1}{273.15 \text{ K}} - \frac{1}{T} \right) \right]$ In this equation, T is the temperature in Kelvin and P is the absolute pressure in the pressure cooker in kPa. Given that the lid of the pressure cooker is a disk with a diameter of 33 cm, what is the force that the lid must be able to withstand at the required pressure?

$$P = \frac{F}{A} \quad \therefore F = P \cdot A$$

$$F = (P_{\text{inside}} - P_{\text{atm}}) \cdot A$$

$$A = \pi r^2$$

$$A = \pi (1.65)^2$$

$$A = 0.855$$

$$T_K = T_C + 273.5$$

$$T_K = 157 + 273.5$$

$$T_K = 430.5$$

ANSWER:

$$40138.21 \text{ N}$$

$$P(T) = 1 \text{ kPa} \cdot \exp \left[(4750 \text{ K}) \left(\frac{1}{273.15 \text{ K}} - \frac{1}{430.5} \right) \right]$$

$$P(T) = 1000 \cdot e^{6.34}$$

$$P = 570222.8 \text{ Pa} \quad \therefore P_{\text{atm}} = 101325$$

$$\therefore F = (570222.8 - 101325) \cdot (0.855)$$

$$F = 40138.21 \text{ N}$$

18. Joshua, a construction worker, holds up a plank of wood 3.2-m long. His left hand pushes the left end of the plank down with a force F_L , while his right hand pushes the plank up with a force F_R at a distance of 0.3 m from the right end of the plank. The plank is held in equilibrium. The plank has a mass of 7 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_L .

$$L = 3.2$$

$$x_{\text{cg}} = \frac{3.2}{2} = 1.6 \text{ m}$$

$$F_L \rightarrow \text{CW}$$

$$F_R \rightarrow \text{CCW}$$

$$W = (m)g \rightarrow (7)(9.8) = 68.6 \text{ N}$$

$$\text{equilibrium} \quad \therefore \tau = 0 \quad \therefore \tau_{\text{net}} = 0$$

$$\therefore \tau_{\text{net}} = 0$$

$$\therefore \tau_{\text{net}} = 0$$

$$\frac{F_R(0.3)}{0.3} = \frac{(1.6)(68.6)}{0.3}$$

$$F_R = 365.8 \text{ N}$$

ANSWER:

$$247.26 \text{ N}$$

$$\text{Upward forces} = \text{downward forces}$$

$$\therefore F_R = F_L + W$$

$$\therefore F_L = F_R - W$$

$$\therefore F_L = 365.8 - 68.6$$

$$F_L = 247.26 \text{ N}$$

+3

19. Aaron drives a 4700-kg truck and descends a 16-m-tall hill at a constant speed. What is the final temperature of 160 kg of brake material if its temperature is 12°C at the top of the hill? Assume the specific heat of the brake material is 830 J/kg°C. Assume the brakes do all the work to maintain the speed.

$$PE_g = mgh$$

$$PE_g = (4700)(9.8)(16)$$

$$PE_g = 736960 \text{ J}$$

$$\text{ANSWER: } 17.55^\circ\text{C}$$

$$\left\{ \begin{array}{l} Q = m_{\text{Brake}} c \Delta T \\ \therefore 736960 = (160)(830) \Delta T \\ \frac{736960}{160 \cdot 830} = \frac{160 \cdot 830}{160 \cdot 830} \Delta T \\ \Delta T = 5.55^\circ \end{array} \right.$$

$$\therefore \Delta T = T_f - T_i$$

$$\therefore T_f = \Delta T + T_i$$

$$T_f = 5.55 + 12 = 17.55^\circ\text{C}$$

20. Daniel is designing a skate park and wants to build a curved ramp (quarterpipe) that leads skaters to a straight ramp. Daniel plans that skaters will drop into the quarterpipe from rest and launch off the straight ramp with a speed of 4.3 m/s. If the height of the quarterpipe is 4.8 m, what must the height of the straight ramp be? Assume that the wheel bearings are well-made and lubricated, so friction can be neglected.

$$\left\{ \begin{array}{l} \text{Quarter Pipe} \\ KE_i \\ PE_g = mgh \\ \therefore \Delta E = W_{nc} \\ \therefore E_f = KE_f + PE_f \end{array} \right. \quad \left\{ \begin{array}{l} \text{Straight ramp} \\ KE = \frac{1}{2}mv^2 \\ PE = mgh \end{array} \right.$$

$$\text{ANSWER: } 3.86 \text{ m}$$

$$\therefore mgh = \frac{1}{2}mv^2 + mgh$$

$$\therefore (4.8)(9.8) = \frac{1}{2}(4.3)^2 + (4.8)h$$

$$\begin{aligned} 47.04 &= 9.245 + 4.8h \\ 37.795 &= 4.8h \\ \frac{37.795}{4.8} &= \frac{4.8h}{4.8} \\ h &= 3.86 \text{ m} \end{aligned}$$

21. Isabella throws a plastic cup horizontally from her dorm window 29.1 m above the ground. Jessica, another student, stands 12.18 m away from the dorm building and catches the plastic cup at a height of 1.87 m above the ground. Jessica claims that Isabella threw the plastic cup too slowly, so Isabella wants to calculate its initial speed to prove her wrong. Ignoring friction, what initial speed does she calculate?

$$\Delta y = 29.1 - 1.87 = 27.23 \text{ m}$$

$$\text{ANSWER: } 5.16 \frac{\text{m}}{\text{s}}$$

$$\Delta v = \frac{\Delta x}{\Delta t}$$

$$\therefore \Delta v = \frac{12.18}{2.36}$$

$$\Delta v = 5.16 \frac{\text{m}}{\text{s}}$$

$$\Delta y = \cancel{20}t + \frac{1}{2}at^2$$

$$-27.23 = \frac{1}{2}(-9.8)t^2$$

$$-27.23 = -4.9t^2$$

$$\sqrt{t^2} = \sqrt{5.557}$$

$$t = 2.36 \text{ s}$$

+3

- ✓ 22. Calculate the work output of a Carnot engine operating between temperatures of 531 K and 256 K for 3430 J of heat transfer to the engine.

ANSWER: 1776.36 J

$$\eta_c = 1 - \frac{T_c}{T_h}$$

$$\therefore \eta = \frac{W}{Q_h}$$

$$\therefore \eta_c = 1 - \frac{256}{531}$$

$$\therefore W = \eta \cdot Q_h$$

$$\therefore W = 0.52189 \cdot 3430$$

$$\eta_c = 0.51289$$

$$W = 1776.36 \text{ J}$$

- 4 23. Charles places a 0.11-kg toy car next to a spring on a track. He moves the car backward to compress the spring 23.9 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it rises to a height of 1.855 cm above the starting position? Since Charles greased the axles of the toy car, assume there is no friction. Charles's previous experiments indicate that the spring has a force constant of 140 N/m.

ANSWER: _____

- 5 24. If a marathon runner averages 95 furlongs per deciday, then how long would it take her to run a 7.7-league marathon? Report your answer in minutes. A furlong is 10 chains, a mile is 1760 yards, a chain is 22 yards, a day is 10 decidays, and a league is 3 miles.

ANSWER: _____

41

25. Grace is trying to keep a 133.3-kg wagon at rest on a frictionless inclined plane, but despite her best efforts, she can only apply a force of 108 N to try to stop the motion. If the plane makes an angle of 23.1° with the horizontal, what is the acceleration of the wagon down the plane?

ANSWER: _____

26. Olivia is a contestant in a winter sporting event. She is tasked with pulling a rope connected to a 980-kg block of ice on a frozen lake. Olivia pulls diagonally upward on the rope at an angle of 55° from the vertical. What is the minimum force with which she must pull in order to get the block of ice moving? Note that for ice on ice, $\mu_s = 0.1$ and $\mu_k = 0.03$.

$F_x \rightarrow F \sin \theta$
 $F_y \rightarrow F \cos \theta$

$S = \mu_s N$
 $\therefore S_{\max} = F \sin \theta$
 $\therefore F \sin \theta = \mu_s (m_s - F \cos \theta)$
 $\therefore F \sin(55) = .1(980 - F \cos 55)$
 $\therefore F \sin 55 + .1 F \cos 55 = 98.0$
 $\therefore F(\sin 55 + .1 \cos 55)$
 $\frac{F(0.8265)}{0.8265} = \frac{98.0}{0.8265}$
 $F = 109.22 \text{ N}$

ANSWER: 109.22 N ✓
 $F \sin \theta = 0$
 $N + F \cos \theta - mg = 0$
 $N = (mg - F \cos \theta)$

27. A typical small rescue helicopter has four blades, each of which is 4.2-m long and has a mass of 47 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1120 kg. If the helicopter's blades rotate at 332 rpm on the ground, to what height could the helicopter rise if all of its blades' rotational kinetic energy is used to lift it?

$I_{\text{blade}} = \frac{1}{3}ML^2 \rightarrow \frac{1}{3}(47)(4.2)^2 = 226.36$
 ANSWER: 60.84 m ✓

There are 4 blades $\therefore 4 \times 226.36 = 1105.44$

$K_{\text{Erot}} = \frac{1}{2}I\omega^2 \rightarrow \frac{1}{2}(1105.44)(34.26)^2$
 $= 662828.14 \text{ J}$

$332 \text{ rpm} \rightarrow \frac{2\pi}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 34.26 \frac{\text{rad}}{\text{s}}$

$K_{\text{Erot}} = mgh$

$\therefore h = \frac{K_{\text{Erot}}}{mh \cdot g}$

$\therefore h = \frac{662828.14}{1120 \cdot 9.8}$

$h = 60.84 \text{ m}$

+3

28. To test the idea of surviving a free-falling elevator, Robert conducts an experiment using a string-attached plank and a programmable toy grasshopper. His makeshift elevator falls 8.8 m and the grasshopper jumps 0.0001 s before impact. Previous tests show that these toys can endure impacts below 6.77 m/s. What is the minimum initial speed needed for the grasshopper's vertical jump to make it impact the ground at a safe speed? (Note: The plank's mass is significantly greater than the toy's mass.)

$$h = 8.8 \text{ m}$$

$$\text{Jump} = 0.0001 \text{ s}$$

$$v_{\text{impact}} = 6.77 \text{ m/s}$$

$$v_F = \sqrt{2gh}$$

$$v_F = \sqrt{2 \cdot 9.8 \cdot 8.8}$$

$$v_F = 13.13 \text{ m/s}$$

$$\text{Impact} = \text{total} + v_{\text{min}}$$

$$v_{\text{min}} = -13.13 + 6.77$$

$$v_{\text{min}} = -6.36 \text{ m/s}$$

ANSWER: 6.36 m/s

$$v = v_0 - gt$$

$$v = (-9.8)(0.0001)$$

$$v = -9.8 \times 10^{-4}$$

$$v_{\text{total}} = -6.36 + (-9.8 \times 10^{-4})$$

$$v_{\text{total}} = -6.36 \text{ m/s}$$

$\therefore v = 6.36 \text{ m/s}$ Vertical Jump

29. Find the increase in entropy as 5.6 kg of ice, originally at 1.0°C, melts to form water at 1.0°C at a particular pressure. The latent heat of fusion for water is 334 kJ/kg.

$$\Delta S = \frac{Q}{T}$$

$$\therefore \Delta S = \frac{1820400}{273.15}$$

$$\Delta S = 6822.54 \frac{\text{J}}{\text{K}}$$

$$Q = mL$$

$$Q = (5.6)(324000)$$

$$Q = 1820400 \text{ J}$$

ANSWER: $6822.54 \frac{\text{J}}{\text{K}}$

$$T_{\text{K}} = T_{\text{C}} + 273.15$$

$$T_{\text{K}} = 1 + 273.15$$

$$T_{\text{K}} = 274.15$$

30. Brandon inflates his bicycle tire to an absolute pressure of 0.75 MPa (a gauge pressure of about 94 psi) at a temperature of 9.1°C. What is the tire's gauge pressure after the temperature has risen to 40.9°C? Assume there are no leaks and the volume of the tire does not change.

$\therefore$ No volume change

$$\therefore P_1 = P_2$$

$$\therefore \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\therefore P_2 = P_1 \left(\frac{T_2}{T_1} \right)$$

$$P_2 = 0.75 \left(\frac{314.05}{282.25} \right)$$

$$P_2 = 0.834$$

ANSWER: 0.233 MPa

$$T_{\text{K}} = 9.1 + 273.15 = 282.25$$

$$T_{\text{K}} = 40.9 + 273.15 = 314.05$$

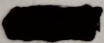
$$\therefore P = P_g + P_{\text{atm}}$$

$$\therefore P_g = P - P_{\text{atm}} = 101325 \text{ Pa}$$

$$\therefore P_g = 0.834 - 0.101$$

$$\therefore P_g = 0.233 \text{ MPa}$$

PHY201 Final Exam on chapter(s): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15

1. -3.15 m/s
2. 5.8 m/s^2
3. 30.21 cm/s
4. 111.62 kJ
5. 3.13 kN
6. 779.19 mmHg
7. 4.2 km
8. 7.9 m/s^2
9. 0.67 s
10. 0.38 rad/s
11. 124.92 kPa
12. 40.93 N
13. 10.4 m/s
14. 902.0 kg/m^3
15. 2.05 m/s^2
16. 5.12 rad/s^2
17. 40.15 kN
18.  297 N
19. $17.55 \text{ }^\circ\text{C}$
20. 3.86 m
21. 5.17 m/s
22. 1776.37 J
23. 850.5 cm/s
24. 280 minutes
25. 3.03 m/s^2
26. 1096 N
27. 60.87 m
28. 6.4 m/s
29. 6.82 kJ/K
30. 733.17 kPa